Basic Network Configuration using Cisco Packet Tracer

Introduction

This project demonstrates a basic network setup using Cisco Packet Tracer. The aim of this project is to understand how devices communicate in a network and how IP addressing works.

Network Design

In this network, I used the following devices:

- 1 Router
- 1 Switch
- 2 End Devices (PCs)

All devices were connected using appropriate cables to form a simple local area network (LAN).

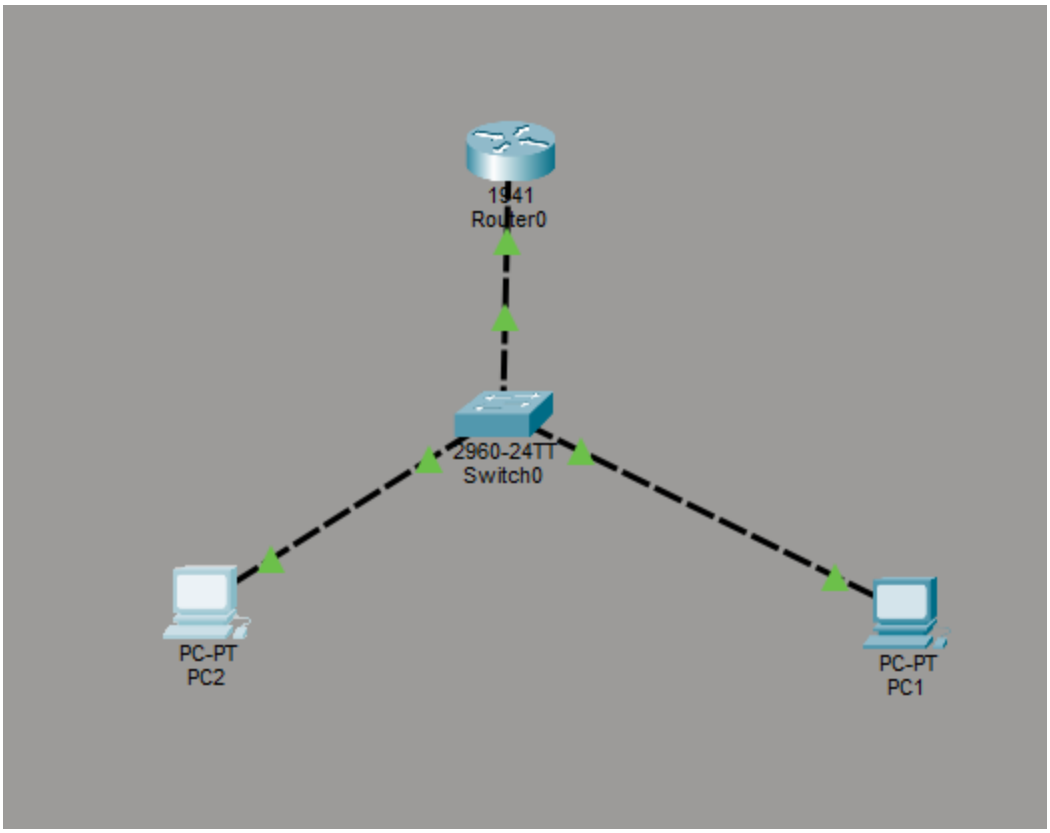
IP Address Configuration

Each device was assigned an IP address manually.

Example:

- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- Router → 192.168.1.1

This allowed all devices to be in the same network and communicate with each other.



⚙️ Configuration Steps

First, I connected all devices using cables in Packet Tracer.

Then, I configured IP addresses on each PC using the desktop IP configuration settings.

Next, I configured the router interface using basic commands in the CLI.

Finally, I saved the configuration and verified the setup.

🔍 Testing Connectivity

To test the network, I used the **ping command**.

I successfully pinged from one PC to another, which confirmed that the network was working correctly.

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=16ms TTL=128
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time=3ms TTL=128
Reply from 192.168.1.3: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 16ms, Average = 6ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=15ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 4ms

C:\>
```

Conclusion

This project helped me understand the basics of networking, including device connections, IP addressing, and communication between systems.

It also improved my practical skills using Cisco Packet Tracer.